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Hydration of hexafluoro acetone on ice

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ABSTRACT

We unravel the fate of hexafluoroacetone (HFA), a PFAS degradation product, on ice by combining surface science and DFT. Temperature programmed desorption (TPD) and X-ray photoelectron spectroscopy (XPS) experimentally show that HFA diffuses into the ice film and readily and completely reacts to form the diol (1,1,1,3,3,3-hexafluoropropane-2,2-diol). Desorption energies were measured for HFA from the underlying substrate (Pt(111)) for the

desorption of the ice film and for the diol as 44, 52 and 62 kJ/mol, respectively. Calculations show that the formation of the diol is favorable on the ice surface by at least 29 kJ/mol. This study is fundamental to better understand the reactivity of HFA on ice in atmospheric conditions since it clearly demonstrates that the diol will dominate reactivity with ice in the environment.

Introduction

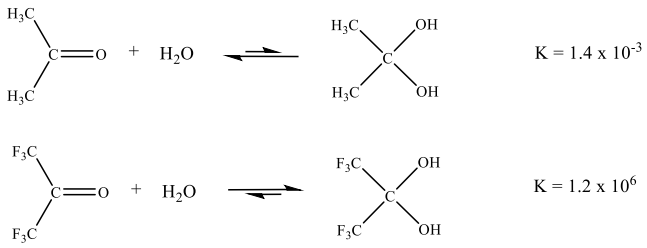
Per- and polyfluoroalkyl substances, PFAS, are ubiquitous in the environment. These compounds, which contain strong carbon-fluorine bonds have been used in a wide range of industrial and consumer products due to their water and oil repellent properties. Their resistance to heat and chemical degradation have allowed them to accumulate over time in the environment. Despite their relative stability, the increasing concentration of these compounds make them a risk not only for health of animals and people, but also as they enter the atmosphere. This is particularly true for smaller PFA molecules that are often produced as secondary emissions from landfill sites, wastewater treatment plants and recycling of PFA polymers.[1,2]

As PFAS enter the atmosphere, interactions with water and ice particles become relevant but they are poorly understood and need to be studied. In this study, we focus on the interactions of hexafluoroacetone (HFA) with ice films in ultra-high-vacuum as a model for ice particles in the atmosphere. Hexafluoroacetone presents itself as a great model for PFAS interactions with ice in the environment because of its simplicity. Yet the de-stabilized carbonyl offers the potential for enhanced reactivity.

HFA is a small perfluorinated molecule analogue to the well-studied acetone molecule. In liquid water, acetone is known to form a tetrahedral hydrate (gem-diol). Electron withdrawal by fluorine

in the vicinity of a carbonyl enhances this tendency by reducing the positive charge on the carbonyl carbon leading to an enhancement of nine orders of magnitude in the equilibrium toward diol formation (scheme 1).[3–6]

Scheme 1. Keto-Diol equilibria



The HFA/ice system has not been previously studied. However, in the analog system acetone/ice, hydrogen bonding dominates the interactions.[7,8] Roberts et al.[9] experimentally showed that amorphous ice, which contains a larger density of dangling OH groups, interacts more strongly with the adsorbate, often leading to more than one thermal desorption state for acetone. Brann et al.[10] found that at low temperatures (<70K), acetone interacts with the dangling bonds in amorphous solid water via hydrogen bonding which produces a shift in the C=O vibration. In fact, hydrogen bonding is the main driving force for interactions between organic moieties and ice.[11] The presence of a diol has not been reported. Besides, the structure of the ice itself is complex and can vary significantly as a function of pressure and temperature. Model ice films have been previously developed.[9,12–17] Under ultra-high vacuum (UHV), three regimes are characteristic, namely porous amorphous solid water, amorphous solid water, and crystalline ice (p-ASW, ASW,

CI, respectively). Obviously, the adsorption of small molecules on ice is tuned by the nature of the ice. Regarding acetone, it was shown that acetone diffuses through the p-ASW but not on the other polymorphs.[10]

From the computational point of view, the HFA-ice system has not been previously investigated. There are, however, numerous studies on the acetone-ice system. In contrast with the available experimental studies, those studies focused on acetone adsorption and dynamics on crystalline ice surfaces (Ih 0001).[7,8,18–20] The surface model is limited even in the case where crystalline ice surfaces are considered since the surface of crystalline ice is poorly characterized.[21] Other ice models were proposed in the literature, where the amorphous nature of the ice surface was obtained by an annealing procedure applied on crystalline ice.[22] An alternative methodology to study these systems is by microsolvation models of the ice-molecule pairs. For example, the hydration of acetone has been previously investigated in detail using a microsolvation approach to model the effect of the H-bond network on the C-O bond formation.[23]

This work focusses on the primary behavior of hexafluoro acetone with porous amorphous ice. We asked whether the strong HFA-water interactions lead to the spontaneous formation of the diol (hexafluoropropane-2,2-diol) even at a low temperature of 120K with solid water. Our observations indicate how favorable this process is, and we explain and quantify this observation.

Experimental and Theoretical Methods

Experiments were carried out in a multi-technique UHV chamber (Specs GmbH) with a base pressure of 4×10^{-10} mbar achieved by a combination of a turbo molecular and ion pumping.

The substrate is an 8 mm diameter and 0.5 mm thick platinum single crystal sample with purity of 99.999% (Princeton scientific). The Pt (111) crystal surface was cleaned by repetitive cycles of 15 min of sputtering (1.5 keV, 1 x 10^{-5} mbar Ar^{+}) at 600K, followed by annealing to 1100 K for 2 minutes. The sample cleanness was confirmed by LEED and XPS. The sample was cooled via a liquid nitrogen reservoir and heated by electron bombardment as needed. The temperature was measured by a chromel-alumel (K-type) thermocouple in contact with the back of the sample.

Hexafluoroacetone (HFA) (97% purity, SynQuest Labs) was introduced to the UHV chamber through a leak valve that was connected to a gas manifold with an HFA lecture bottle. Exposure of HFA to the surface was typically performed at $1x10^{-7}$ Torr pressure and the dosage measured in Langmuirs (1L = $1x10^{-6}$ Torr.s). Doubly deionized water was introduced to the UHV chamber through a leak valve attached to a dedicated manifold. Water exposure was performed at $5x10^{-7}$ Torr.

Temperature Programmed Desorption

The TPD experiments require placing the sample in front of a differentially pumped quadrupole mass spectrometer (QMS, Pfeiffer Prisma) and ramping the sample temperature while monitoring the fragments of interest. For our experiments, the Pt(111) sample was dosed with water and/or varying exposures of HFA at 120 K and placed 1 mm below a 1 mm diameter hole leading to the differentially pumped QMS. The temperature-controlled heating ramp was programmed in LabView and designed to provide and record a linear temperature ramp in the range of 120 K to 600 K, with a ramp rate of 4 K/s. The sample was subject to one cleaning cycle (Ar sputtering followed by annealing to 1100 K) to ensure the cleanness of the surface before experiments. A

total of 10 to 12 channels were monitored for selected m/z values corresponding to the main peaks of the fragmentation pattern of the expected species.

X-ray Photoelectron Spectroscopy

XPS spectra were recorded on a Specs GmbH system using a XR50 X-ray source and a Phoibos 100 SCD analyzer. A standard Al Kα source (1486.7 eV) was operated at 380 W (14.6 kV, 26 mA). Low resolution surveys were always obtained, and selected peaks were obtained in high-resolution spectra using 0.1 eV step size, 0.5 second dwell time, and a pass energy of 30 eV. For the C1s and O1s regions 8 and 6 scans respectively were acquired to increase the signal to noise ratio. The spectra were then fit using CasaXPS analysis software using a mixed Gaussian-Lorentzian function and Shirley background subtractions for the C1s, F1s and Pt4f regions while a linear background subtraction was used for the O1s region. The energy scale is calibrated using adventitious carbon at 285.0 eV.

For experiments on clean Pt(111), HFA was dosed on the clean sample at <120K and then heated as required to the desired temperature of observation to acquire the spectrum. Once the acquisition was done, the sample was flashed to 1100K to remove adsorbed species. For experiments on ice films, water was dosed on the clean sample at <120K, the film was typically scanned for cleanliness before HFA dosing. Once measurements were done, the sample was flashed to 1100K to remove adsorbed species. This process was repeated for each dose of interest.

Computational details

Microsolvation calculations were performed using Gaussian[24] at the M062X/def2TZVP level,[25–27] and PCM correction for water.[28] For every structure study a variety of possible configurations for the solvating water molecules were computed, in order to locate the most stable structures.

Periodic calculations were performed using CP2K[29] at the PBE-D3/DZVP level.[30–33] The amorphous ice surface was obtained from an annealing procedure that started with a (3x3x1) supercell of tetragonal ice. Cell parameters were obtained by bulk optimisation and the surface that was used was the 0001 surface with 4 bilayers and 9 Å of vacuum above the surface resulting in an 18.98 Å x 20.33 Å x 20 Å cell. The amorphous surface was obtained by subjecting a crystalline ice surface to a free MD simulation for 20 ps at 330 K. Fifteen low energy structures were chosen from the simulation and optimized, the most stable of which was chosen as a model amorphous ice surface. The corresponding structure is provided in supporting material.

The XPS C 1s binding energies were computed ab initio using the Vienna Ab-Initio Simulation Package (VASP)[34] with the PBE functional[30] and dDSc dispersion correction.[35] Ion-electron interactions were treated using projector augmented wave (PAW)[36] formalism with 645 eV cutoff for the augmentation charge. The plane wave basis set was truncated at 400 eV. Calculations were performed at the Γ-point. The XPS calculations employed the final-state approximation by exciting half of an electron to a virtual orbital. The excited electrons are then allowed to relax upon removal of the core hole, and all nonexcited core electrons remain frozen. Each atom is excited in an independent calculation. Only the relative values of core level binding energies obtained through this method are reliable.[37–39]

Results and Discussion

The model for ice particles in the atmosphere was generated by creating a film of ice on a Pt(111) surface. Three different types of ice film are generated depending on the surface temperature in UHV.[10] Below 135 K an amorphous porous ice film is produced, between 135 and 150 K, the amorphous film densifies, and above 150 K a crystalline ice layer is expected. As this is a dynamic system, a surface temperature of ~120 K was chosen to reproducibly obtain the amorphous porous ice film, also referred to as porous amorphous solid water (p-ASW). This film was chosen for its simplicity of preparation and reproducibility, but also because it is expected to be most reactive in interactions with adsorbates.[9,10]

Enough water was dosed into the UHV system at a surface temperature of ~120 K to produce an ice layer sufficiently thick for the surface of the ice to be unaffected by the underlying substrate. A dose of 250 L of water dosed at 120K was used as our reference ice film with a thickness of 13 nm, as estimated from the attenuation of the Pt 4f peaks in XPS. To determine this thickness, the inelastic mean free path (IMFP) for ice was calculated as 4.5 nm using the NIST database.[40]

However, since this most-reproducible ice film is porous, a test was performed to determine the adsorption of HFA on ice films of different thicknesses. Figure 1 shows the TPD results from the deposition of HFA over different doses of water. Multiple masses were monitored, as the fragmentation patterns of the parent HFA and potential reaction products are complex and overlapping. The figure shows the expected desorption of water (m/z=18) and the desorption of the parent HFA molecule (m/z=166). It also clearly shows a peak that is characteristic of the diol[41] ($CF_3C(OH)_2CF_3$) at m/z=45, which is assigned to a COOH fragment. The parent diol does not produce a molecular ion (m/z=184) in the mass spectrometer, and out of the fragments produced, m/z=45 is not present in the mass spectrum of HFA,[42] has a reasonable intensity and is clearly identifiable. For increasing amounts of water, a desorption peak is observed at 184 K for both

masses (m/z=166 and m/z=45). The source of this peak is ascribed to the desorption of the diol molecule. The experiment cannot distinguish if the presence of m/z=166 is related to the diol or to unreacted HFA.

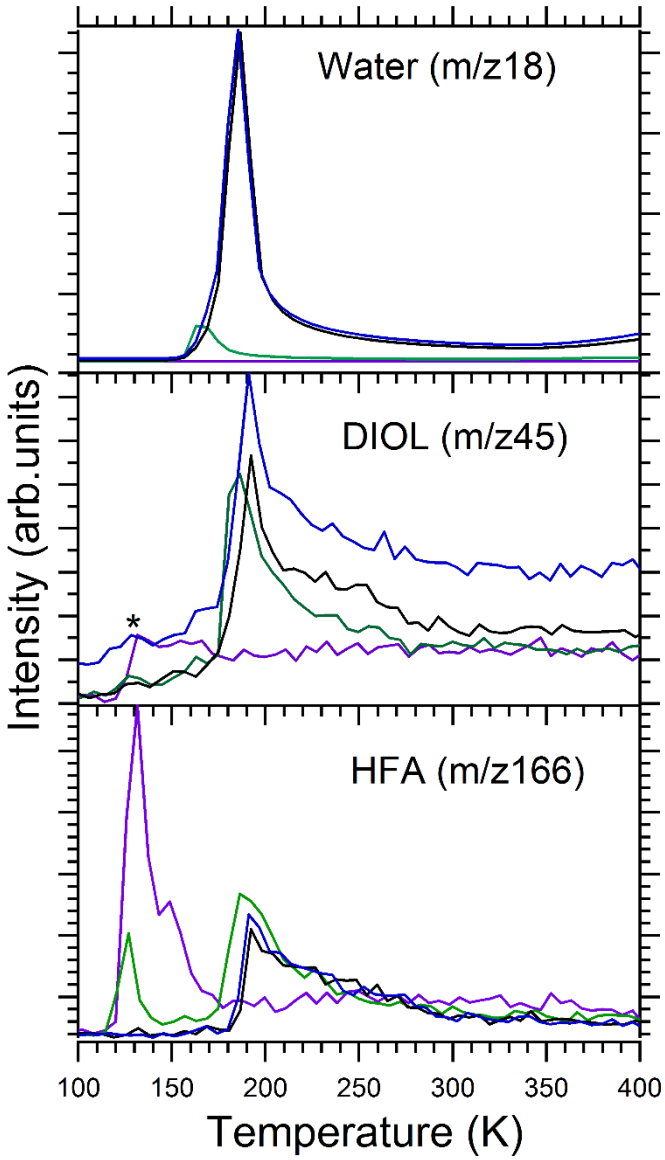


Figure 1. TPD spectra of 5 Langmuir doses of HFA deposited on increasing thickness of water on top of Pt(111) substrate at 120 K. Purple: 5L HFA/Pt(111); green: 5L HFA/25L H_2O/Pt(111); black: 5L HFA/250L H_2O/Pt(111). An inverse dose of water on top of HFA was also done, blue:

250L H_2O/ 5L HFA/Pt(111). * indicates an experimental artifact, namely a jump in the baseline for the black line.

Desorption of water from Pt(111) has been previously studied.[9,16,43,44] Our data in Figure 1 is consistent with literature, although we focus on ice films that are much thicker. The desorption temperature of the ice film has a maximum at 178K for a 250L dose. On the other end, HFA desorbs from clean platinum at much lower temperature. Figure 1 shows a 5 L HFA dosed on increasingly thick ice films, starting on clean Pt(111). TPD spectra from the clean Pt(111) produces sharp peaks at 131K, in agreement with Avery,[45] in which a monolayer peak is observed at 120K, while a lower temperature multilayer peak is only expected at T<110K. These observations were corroborated by measuring the TPD spectra for different doses of HFA on clean Pt(111). Figure S1 shows that a dose of 7L HFA in our chamber corresponds to a monolayer on Pt(111). As expected, HFA/Pt(111) does not produce any m/z=45, characteristic of the diol.

While it was expected that HFA would "sit on top" of the ice at 120K as seems to be the case for acetone,[8,18–21] the results suggested the possibility of molecular diffusion of HFA into the ice film. The progression from the clean platinum to the 250 L thick ice film is shown by the reduction of the HFA desorption peak from Pt(111) at 131K, until its minimization at 250 L of water. The observation of HFA desorption from Pt(111) in the presence of the ice film indicates water clustering allowing portions of the underlying surface to remain exposed until sufficient water is present. The maximum of the broad peak of the diol, at 184K, shifts and desorption seems to take place after the water desorption peak. This then suggests the diffusion through the underlying water film which would allow for the diol's subsequent desorption. In order to probe this diffusion

interaction between the HFA and ice film, a reverse experiment was performed, that is a dose of 5 L of HFA was dosed first on clean Pt(111), followed by 250L of water. While there are slight changes in intensity, especially for m/z=45, the resulting TPD spectra does not seem to change from the normally dosed set of masses, further indicating diffusion and diol formation within the film. In the system acetone/ice, acetone has been seen to diffuse when the ice layer is porous (p-ASW).[10]

Increasing amounts of HFA dosed on the thick ice film (250L water) shows growth of the peak m/z=166, which does not saturate well above the usual dose required for a monolayer of HFA (7L), Figure 2. This reinforces the idea that HFA diffuses into the ice film and reacts to make the diol. Interestingly, the peak characteristic of the diol does not increase to the same extent. This could indicate the co-existence of the keto and diol forms within the ice film leading to saturation of the diol available sites, and/or the loss of water from the diol during heating and desorption (ie. diol decomposition into HFA between the surface and the detector).

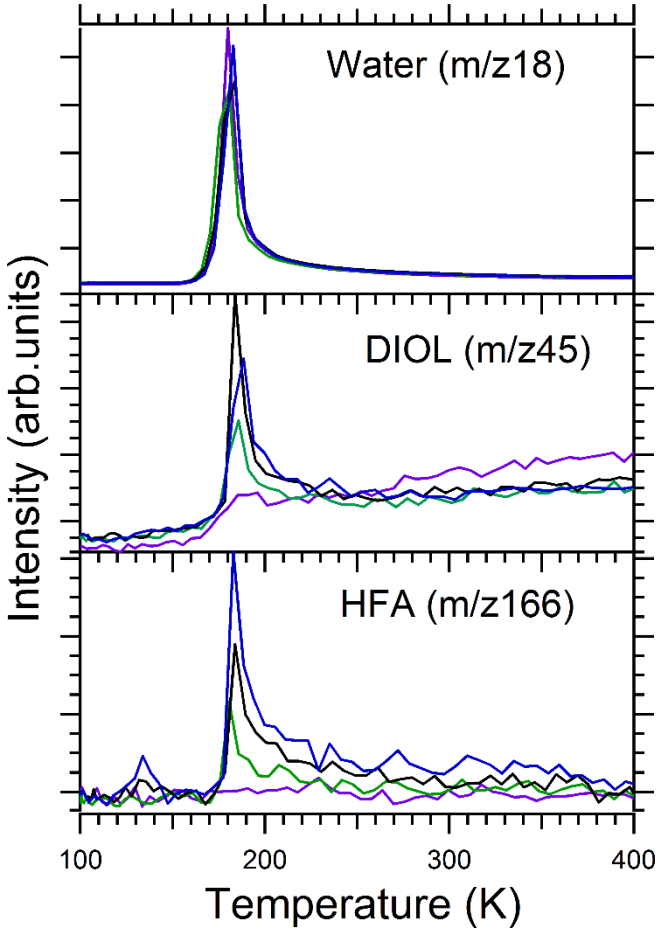


Figure 2. TPD spectra of increasing doses of HFA deposited on 250L water on Pt(111) substrate at 120K. Main desorption peak appears at 184K for all doses. Purple: 1L HFA/250L H_2O/Pt(111); green: 5L HFA/250L H_2O/Pt(111); black: 10L HFA/250L H_2O/Pt(111); blue: 20L HFA/250L H_2O/Pt(111).

Using the TPD data, the corresponding desorption energies ($\Delta_{des}E$) were calculated (Table 1). The approach taken is based on the general rate law for desorption, the Polanyi-Wigner equation, and using the simple Redhead method (equation 1). The frequency factors (ν) were calculated using the method by Campbell, et.al [46,47] in which the frequency factor for molecular desorption is

corrected based on known standard entropy values for the adsorbed and gas phase molecules obtained from literature.[48]

$$\frac{\Delta_{des}E}{RT_p^2} = \frac{n\nu\theta^{n-1}}{dT/dt}\exp\left(-\frac{\Delta_{des}E}{RT}\right) \qquad \text{(eq. 1)}$$

The desorption energy values obtained improve on previously proposed quantities. For example Avery[45] determined the desorption energy for the first layer of HFA on Pt(111) as 35 kJ/mol using the rough approximation of 10^{13} for the frequency factor. The authors found the same heat of desorption for the multilayer, although the multilayer desorption temperature was measured at ~100K and they suggest the value of 35 kJ/mol for the first layer is an underestimation. Indeed, we obtain a value of 44 kJ/mol for the first layer desorption energy.

Table1. Desorption energies from Pt(111).

Mass (m/z)	Molecule	Temperature (K)	ν (s^{-1})	$\Delta_{des}E$ (kJ/mol)	$\Delta_{des}E$ (lit. in kJ/mol)
166	HFA	131	3.4 x 10^{17}	44	35[45]
18	water	178	1.4 x 10^{15}	52	<47 -55[#9] 47.2[#43] 49.4[&49]
45	Diol/water/Pt	184	4.4 x 10^{17}	62*	

* This is an apparent desorption energy, since the diol is embedded in the ice film, see text.

The desorption energy varies from ~55 kJ/mol at the monolayer to 47 kJ/mol at the multilayer. Campbell[43] reports the value for an amorphous ice film.

& Ice sublimation energy

Although the diol desorption temperature is higher than that of water, the temperature difference between the desorption peaks of water and the diol is small, yet the measured desorption energy for the diol is ~10 kJ/mol higher. This may be explained by the diol desorbing together with the water, or in fact water desorbing preferentially.

XPS was used to monitor the accumulation of HFA following an increasing dose of HFA on the ice film (250L water). As shown in Figure 3, the C1s region produces 2 indicative peaks of HFA, one for the carbonyl carbon, labelled C-O and one for the fluorinated methyl carbons (C-F). These peaks appear at 289.4 and 293.0 eV, respectively, well within the range of the expected positions.[50] On clean platinum, where formation of the diol is not possible, the two peak positions for HFA are 286.7 and 293.3 eV (Figure 4), indicating the presence of the diol in the ice film. This shift in the C 1s binding energy upon hydration of HFA was confirmed by DFT modeling (see below).

A third peak is also seen due to accumulation of adventitious carbon on the ice film. The indicative peaks of HFA continue to grow as increasing doses of HFA are accumulated onto the ice film at 120K. As indicated from the TPD data, the large accumulation of HFA corresponds to diffusion of HFA into the film rather than multilayer formation.

The oxygen region of the photoelectron spectrum is dominated by the water signal which is 1-2 orders of magnitude larger. Oxygen arising from HFA/diol adsorbate cannot be isolated. The F1s region shows a single peak at 688.2 eV. Both spectra are presented in Figure 3.

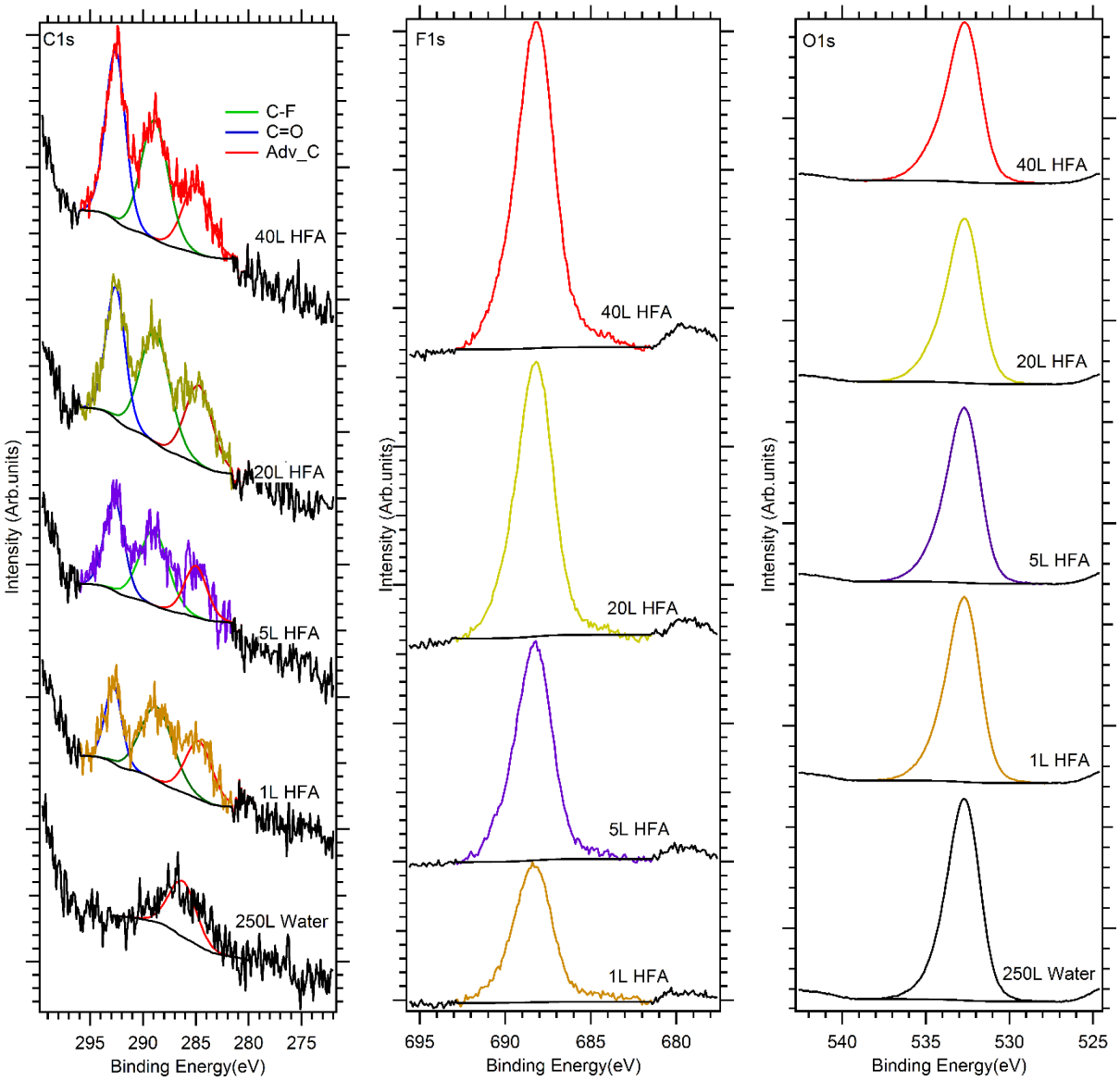


Figure 3. C1s, F1s and O1s XPS region spectra of increasing amounts of HFA dosed on top of 250 L of water prepared on Pt(111) substrate at 120 K. Peaks for C1s corresponding to C-F, C-O, and adventitious carbon were fitted with binding energies of 292.6, 289.0, and 285.0 eV, respectively.

Formation of the diol in the presence of water is energetically favourable as discussed below. In addition, a diolate has been previously observed in surface reactions when the surface provides a strong enough available reactive oxygen, as in some oxide surfaces. For example, the fraction of diolate is controlled by the degree of reduction in a rutile surface.[51,52]

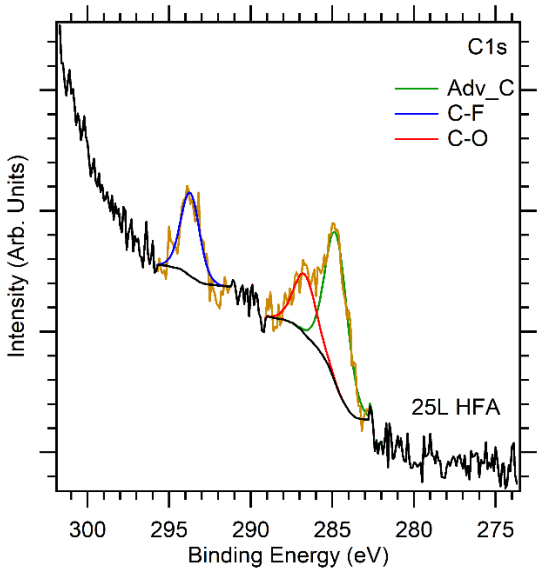


Figure 4. X-ray photoelectron spectra for the C1s region for a saturated layer of HFA onto Pt(111).Although the Pt substrate was clean at the beginning of the experiment, adventitious carbon is present due to accumulation of carbon during experiments.

We also employed a computational approach in order to corroborate which species are present on or in the ice surface, in particular if the condensation reaction (HFA + H_2O to produce the diol, Scheme 1) is thermodynamically favoured under these conditions. As it is difficult to produce a realistic computational model of the amorphous-porous ice we approached the problem from two opposite directions.

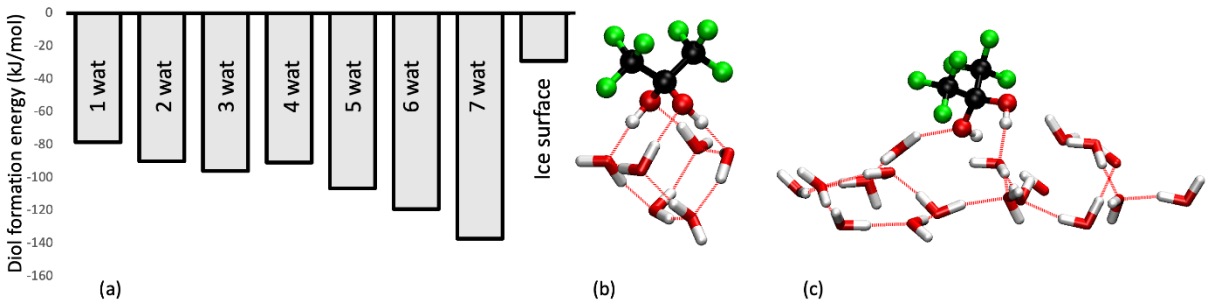


Figure 5. (a) Reaction energy for HFA + H2O → diol for varying numbers of water molecules. The first seven energy values come from microsolvation simulations with a given number of water molecules (the number given is the number with the HFA before the reaction), the last value comes from a periodic calculation with an amorphous ice surface. (b) Diol/water interaction modelled using microsolvation. This is the structure obtained from the reaction of the HFA in presence of seven water molecules. (c) Diol adsorbed on an amorphous ice surface. Hydrogen bonds are represented up to distance of 3.0 Å and angle of 32°.

With a first set of calculations, we studied the condensation reaction in microsolvation conditions, thus reproducing the situation where the local structure of the amorphous-porous ice would be entirely determined by the presence of the HFA or the diol. After conducting the microsolvation condensation reaction with varying numbers of water molecules (1-7), it was shown that the condensation reaction is always thermodynamically favourable, as can be seen in Figure 5a, which shows the condensation energies for different numbers of water molecules. The reaction energy varies from -79 kJ/mol when only one water molecule is present (and therefore no water molecules are present to interact with the diol after the reaction) to -137 kJ/mol when seven water molecules are present (the resulting structure with the diol interacting with six water

molecules is shown in Figure 5b). This range compares favourably with the experimental value determined by solvation and subsequent Born-Haber analysis, which finds the standard enthalpy change for this reaction in solution as -93.72 kJ/mol at 298K.[53]

We also performed a second set of calculations that reproduce an amorphous ice surface in periodic boundary conditions. When adsorbed on this model amorphous surface (Figure 5c), the condensation reaction from HFA to diol was still favourable (-29 kJ/mol).

While neither of these computational approaches fully reproduces the condensation reaction on or in amorphous-porous ice, they represent two extreme cases, one where the adsorbed molecule locally rearranges the ice structure to increase adsorption energy, the other where the molecules are adsorbed on a surface that has little opportunity to rearrange itself so as to better adsorb them.

Having studied these two extreme cases and shown that in each one the condensation reaction from HFA to diol is thermodynamically favoured, the computational approach shows that HFA is spontaneously hydrated into the diol in presence of ice.

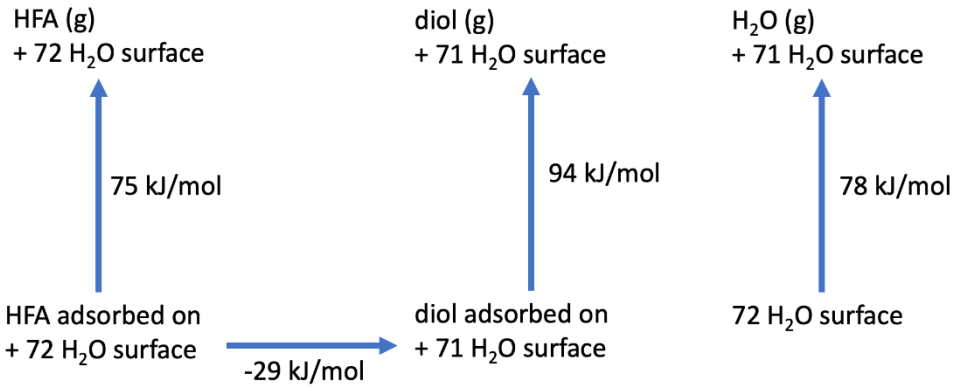


Figure 6. Thermodynamic cycle for the adsorption and reactivity of HFA and the diol on the model periodic amorphous ice surface.

Figure 6 shows the adsorption and reaction energies of HFA and the diol on the model periodic amorphous ice surface. The adsorption energies were calculated on the periodic ice surface only as microsolvation is not an appropriate method for doing so due to the presence of magic numbers of water molecules that can create particularly stable hydrogen bonding networks (this can be seen for example in the decease of the reaction energy between 3 and 4 water molecules in Figure 4a). The resulting desorption energies are systematically greater than those obtained experimentally, a difference that is due to the lack of a large-scale reorganization of the water surface in the calculations after desorption. It is however significant that the difference in desorption energies between water and the diol is of 16 kJ/mol a value similar to the 10 kJ/mol measured experimentally (a variation of 6 kJ/mol is within the expected margin of error for DFT calculations), while the difference is only 3 kJ/mol for HFA desorption. This gives clear evidence that it is the diol that is detected in the ice and not HFA.

Calculating the XPS binding energies gives confirmation that it is indeed the diol that is present on the surface. For the diol, the gap between the two C1s binding energies (central carbon and CF_3 carbon) is 3.8 eV, which closely matches the 3.6 eV from experimental measurements (Figure 3). For HFA the gap is larger (calculated at 4.8 eV), in qualitative agreement with the larger gap observed for HFA on clean Pt(111) in Figure 4.

Conclusions

Both acetone and hexafluoroacetone (HFA) are known to react with water to form the corresponding diol species, with HFA being much more reactive. Considering the interaction of these molecules with ice films as models for ice particles in the atmosphere, acetone has been

previously shown to interact by H-bond formation. In contrast, we show here that HFA diffuses into the ice layers of porous amorphous solid water (p-ASW) and readily forms the diol. We have shown that this process is favorable at ice surfaces by a minimum of -29 kJ/mol. Once the diol is formed, the films can be heated and we determined that sublimation of water competes and in fact starts before the elimination of the diol from the ice. This study demonstrates that the diol will dominate HFA reactivity with ice in the environment.

Supporting Information.

TPD spectra of HFA/Pt(1101), XPS data for F1s and O1s regions of HFA/ice/Pt(111), XPS data for C1s region of HFA/Pt(111), IRAS spectrum of (nominal) HFA/ice/Pt(111), and cartesian coordinates for calculated structures (Pdf).

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Notes

The authors declare no competing financial interest.

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Graphical Abstract

